

Course Code: CST413

Course Name: MACHINE LEARNING

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- 1 Describe the concept of semi- supervised Learning and list out any two applications of Semi-Supervised Learning. (3)
- 2 Describe reinforcement learning with example (3)
- 3 Discuss logistic Regression. Write the pros and cons of logistic regression. (3)
- 4 Write short notes on the gradient descent method? Find its usage in the backpropagation algorithm? (3)
- 5 Calculate the output Y of three input neuron with bias. The input feature vector is $(x_1, x_2, x_3) = (0.8, 0.6, 0.4)$ and weight values are $[w_1, w_2, w_3, b] = [0.1, 0.3, -0.2, 0.35]$. Use binary Sigmoid as activation function. (3)
- 6 Show that the function (3)

$$K(\vec{x}, \vec{y}) = (\vec{x} \cdot \vec{y})^2.$$
 is a kernel function
- 7 Given the following data, Find the covariance matrix of the given table. (3)

x	2	1	0	-1
y	4	3	1	0.5
- 8 Calculate (3)
 - a) The Levenshtein distance between “HORSE” and “ROS”
 - b) The Euclidian distance and Manhattan Distance between two points A (1,2,3,1) and B (2,1,2,3).
- 9 Describe cross-validation in machine learning? Also describe leave-one-out cross validation. (3)
- 10 Discuss Bootstrapping in machine learning (3)

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) Discuss “learning” in the context of machine learning? Describe the basic components of the machine learning process. (14)

OR

- 12 a) Write short notes on classification problem in machine learning? Give any five examples of classification problems from real life situations. (7)
- b) A coin is tossed 100 times and lands heads 55 times. Find the maximum likelihood estimate for θ , the probability of heads? (7)

Module II

- 13 a) The following table contains factors that determines whether the species belongs to M or H. Use naive Bayes algorithm, to classify a particular species to M or H, having color=green, legs=2, Height=tall and smelly=no. (7)

No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

- b) Given the following data, construct a linear regression model. (7)

X	2	4	6	8
Y	3	7	5	10

OR

- 14 a) Identify the first splitting attribute for decision tree by using ID3 algorithm with the following dataset. (10)

a1	a2	a3	Classification
True	Hot	High	No
True	Hot	High	No
False	Hot	High	Yes
False	Cool	Normal	Yes
False	Cool	Normal	Yes
True	Cool	High	No
True	Hot	High	No
True	Hot	Normal	Yes
False	Cool	Normal	Yes
False	Cool	High	Yes

- b) Discuss overfitting of data with an example. (4)

Module III

- 15 a) State the mathematical formulation of the SVM problem. Give an outline of the method for solving the problem (7)
- b) Describe the basic outline of the backpropagation algorithm? (7)

OR

- 16 a) Implement AND function using perceptron network for the given bipolar input (X1, X2) and target(T), where learning parameter is 1. (7)

X1	X2	T
1	1	1
1	-1	-1
-1	1	-1
-1	-1	-1

- b) State kernel function? List out any five important kernel function. (7)

Module IV

- 17 a) Use K Means clustering to cluster the following data into 3 groups (10)
 $A1=(2,10)$, $A2=(2,5)$, $A3=(8,4)$, $A4=(5,8)$, $A5=(7,5)$, $A6=(6,4)$, $A7=(1,2)$, $A8=(4,9)$. Suppose that the initial seeds (centers of each cluster) are A1, A4 and A7.
- b) Describe the steps involved in Expectation Maximisation Algorithm. (4)

OR

- 18 a) Given the following distance matrix, construct the dendrogram using agglomerative clustering with average linkage. (10)

Item	A	B	C	D	E
A	0	2	3	3	4
B	2	0	3	5	4
C	3	3	0	2	6
D	3	5	2	0	4
E	4	4	6	4	0

- b) Summarize dimensionality reduction? Analyze the usefulness of dimensionality reduction in machine learning. (4)

Module V

- 19 a) Demonstrate the random forests method used in machine learning. (7)
- b) Suppose 10000 patients get tested for flu; out of them, 9000 are actually healthy and 1000 are actually sick. For the sick people, a test was positive for 620 and negative for 380. For the healthy people, the same test was positive for 180 and negative for 8820. Construct a confusion matrix for the data and compute the accuracy, precision and recall for the data. (7)

OR

- 20 a) Interpret ROC space and ROC curve in machine learning? Given the following data, construct the ROC curve of the data and mark the AUC. (14)

Threshold	TP	TN	FP	FN
1	0	25	0	29
2	7	25	0	22
3	18	24	1	11
4	26	20	5	3
5	29	11	14	0
6	29	0	25	0
7	29	0	25	0
